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# Cambridge O Level

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**MATHEMATICS (SYLLABUS D)** **4024/12**

Paper 1 Non-calculator **May/June 2026**

**2 hours**

You must answer on the question paper.

You will need: Geometrical instruments

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.
- You may use tracing paper.
- You must show all necessary working clearly.

**INFORMATION**

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.

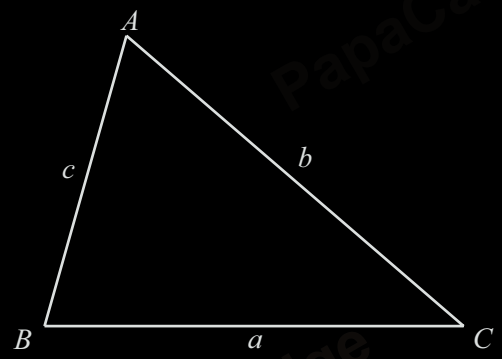
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**[Turn over**

## List of formulas

| | |
|---|---|
| Area, $A$, of triangle, base $b$, height $h$. | $A = \frac{1}{2}bh$ |
| Area, $A$, of circle of radius $r$. | $A = \pi r^2$ |
| Circumference, $C$, of circle of radius $r$. | $C = 2\pi r$ |
| Curved surface area, $A$, of cylinder of radius $r$, height $h$. | $A = 2\pi rh$ |
| Curved surface area, $A$, of cone of radius $r$, sloping edge $l$. | $A = \pi rl$ |
| Surface area, $A$, of sphere of radius $r$. | $A = 4\pi r^2$ |
| Volume, $V$, of prism, cross-sectional area $A$, length $l$. | $V = Al$ |
| Volume, $V$, of pyramid, base area $A$, height $h$. | $V = \frac{1}{3}Ah$ |
| Volume, $V$, of cylinder of radius $r$, height $h$. | $V = \pi r^2 h$ |
| Volume, $V$, of cone of radius $r$, height $h$. | $V = \frac{1}{3}\pi r^2 h$ |
| Volume, $V$, of sphere of radius $r$. | $V = \frac{4}{3}\pi r^3$ |
| For the equation $ax^2 + bx + c = 0$, where $a \neq 0$ | $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ |

For the triangle shown,




$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$\text{Area} = \frac{1}{2}ab\sin C$$

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Calculators must **not** be used in this paper.

**1** Write down the value of

**(a)** $5^3$

..... [1]

**(b)** $\sqrt{121}$.

..... [1]

**2** Buses travel from the city centre to the airport.

**(a)** A bus leaves the city centre at 10 05.
The bus journey from the city centre to the airport takes 1 hour 25 minutes.

Work out the time the bus arrives at the airport.

..... [1]

**(b)** Another bus arrives at the airport at 16 20.
This bus journey from the city centre to the airport takes 1 hour 35 minutes.

Work out the time this bus leaves the city centre.

..... [1]

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**3** These are the scores for each of 10 children in a spelling test.

7 10 10 15 7 5 11 7 13 5

**(a)** Work out the median.

..... [2]

**(b)** Work out the mean.

..... [2]

**4** Ana has 60 cards.
Each card is red or green.

**(a)** The ratio red cards : green cards = 2 : 1.

Show that there are 20 green cards.

[1]

**(b)** Each of the 60 cards is a square or a triangle.
The ratio square cards : triangle cards = 5 : 1.

There are 16 green square cards.

Complete the two-way table.

| | **red** | **green** | **total** |
|---|---|---|---|
| **square** | | 16 | |
| **triangle** | | | |
| **total** | | 20 | 60 |

[2]

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**5** The scale drawing shows the positions of two villages, *A* and *B*.
The scale is 2 cm to 1 km.




scale: 2 cm to 1 km

Farm *C* is 4 km from village *A* and 2.5 km from village *B*.
Farm *C* is further south than village *B*.

**(a)** **Using a ruler and compasses only**, construct the position of farm *C*. [3]

**(b)** Measure the bearing of farm *C* from village *A*.

..... [1]

**6** Pran invests $2000 in a savings account.
The account pays simple interest at a rate of 3.5% per year.

Calculate the value of the investment at the end of 4 years.

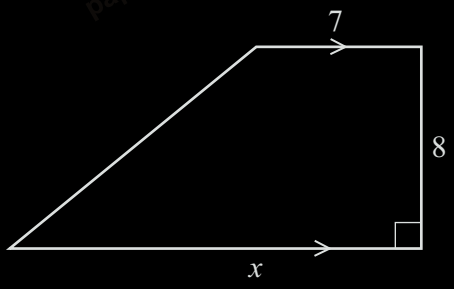
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**7** The diagram shows a trapezium.
All the lengths are in centimetres.




NOT TO SCALE

The area of the trapezium is $100\,cm^2$.

Work out the value of $x$.

$x =$ ..... [3]

**8** Leo has a 5-sided spinner numbered from 1 to 5.
Leo spins the spinner 500 times.
His results are shown in the table.

| **number on spinner** | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| **frequency** | 105 | 96 | 95 | 104 | 100 |

**(a)** Leo says that the results show that the spinner is fair.

Explain why he is correct.

..... [1]

**(b)** Find the relative frequency of the spinner landing on 5.

..... [1]

**(c)** Mary spins this spinner 100 times.

Find the number of times Mary should expect the spinner to land on 5.

..... [1]

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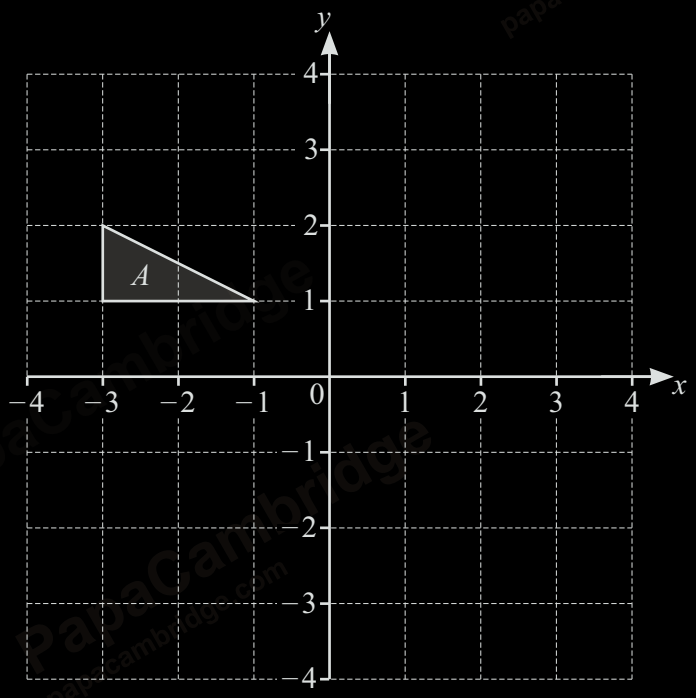
**9** Work out.

$$1\frac{2}{3} \times \frac{9}{10} + 2\frac{1}{5}$$

Give your answer as a mixed number in its simplest form.

..... [4]

**10** The diagram shows triangle $A$.




Reflect triangle $A$ in the line $y = x$. [2]

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**11** **(a)** Write 400 as a product of its prime factors.

..... [2]

**(b)** Find the highest common factor (HCF) of 400 and 180.

..... [2]

**12** Rearrange the formula to make $m$ the subject.

$k = 4(m - p)$

$m =$ ..... [2]

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**13** **(a)** The $n$th term of a sequence is $n+8$.

Find the first three terms of this sequence.

..... , ..... , ..... [1]

**(b)** These are the first four terms of a different sequence.

$T_1 = 4$ $T_2 = 10$ $T_3 = 16$ $T_4 = 22$

**(i)** Write down the term-to-term rule for this sequence.

..... [1]

**(ii)** Find an expression for the $n$th term of the sequence, $T$.

$T_n =$ ..... [2]

**(c)** The $n$th term of another sequence is $R_n = \dfrac{T_n}{n+8}$.

The $k$th term of this sequence is 5.

Find the value of $k$.

$k =$ ..... [3]

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**14** **(a)** Write $350\,000$ in standard form.

..... [1]

**(b)** Write $2.7 \times 10^{-4}$ as an ordinary number.

..... [1]

**(c)** Work out.

$$1.3 \times 10^{12} + 4.5 \times 10^{13}$$

Give your answer in standard form.

..... [2]

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**15** The equation of line $L$ is $2y + x = 19$ .

**(a) (i)** Find the gradient of line $L$.

..... [2]

**(ii)** Find the $y$-intercept of line $L$.

..... [1]

**(b)** Line $M$ is perpendicular to line $L$.
Line $M$ passes through the point $(-1, -5)$.

Show that the equation of line $M$ is $y = 2x - 3$ .

[2]

**(c)** Find the coordinates of the point of intersection of the lines $2y + x = 19$ and $y = 2x - 3$ .

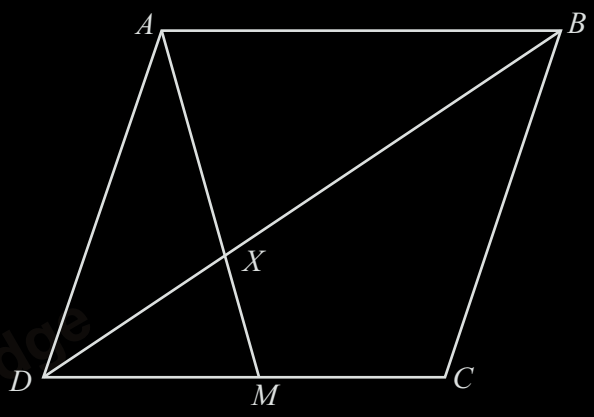
( ..... , ..... ) [3]

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**16**




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*ABCD* is a rhombus.
*M* is a point on *DC*.
*AM* and *BD* intersect at *X*.

**(a)** Show that triangle *AXB* is similar to triangle *MXD*.
Give a geometrical reason for each statement you make.

.....

.....

.....

..... [3]

**(b)** *M* is the midpoint of *DC*.
*AB* = 8 cm and *BD* = 15 cm.

**(i)** Calculate *DX*.

*DX* = ..... cm [2]

**(ii)** Complete the statement.

area of triangle *AXB* : area of triangle *MXD* = ..... : 1 [1]

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**17** In a right-angled triangle

- the base is $x$ cm
- the height is 7 cm shorter than the base
- the hypotenuse (longest side) is 2 cm longer than the base.

**(a)** Use this information to form an equation and show that it simplifies to $x^2 - 18x + 45 = 0$.

[4]

**(b)** Solve by factorising.

$$x^2 - 18x + 45 = 0$$

$x =$ ..... or $x =$ ..... [3]

**(c)** Find the area of the triangle.

..... $cm^2$ [2]

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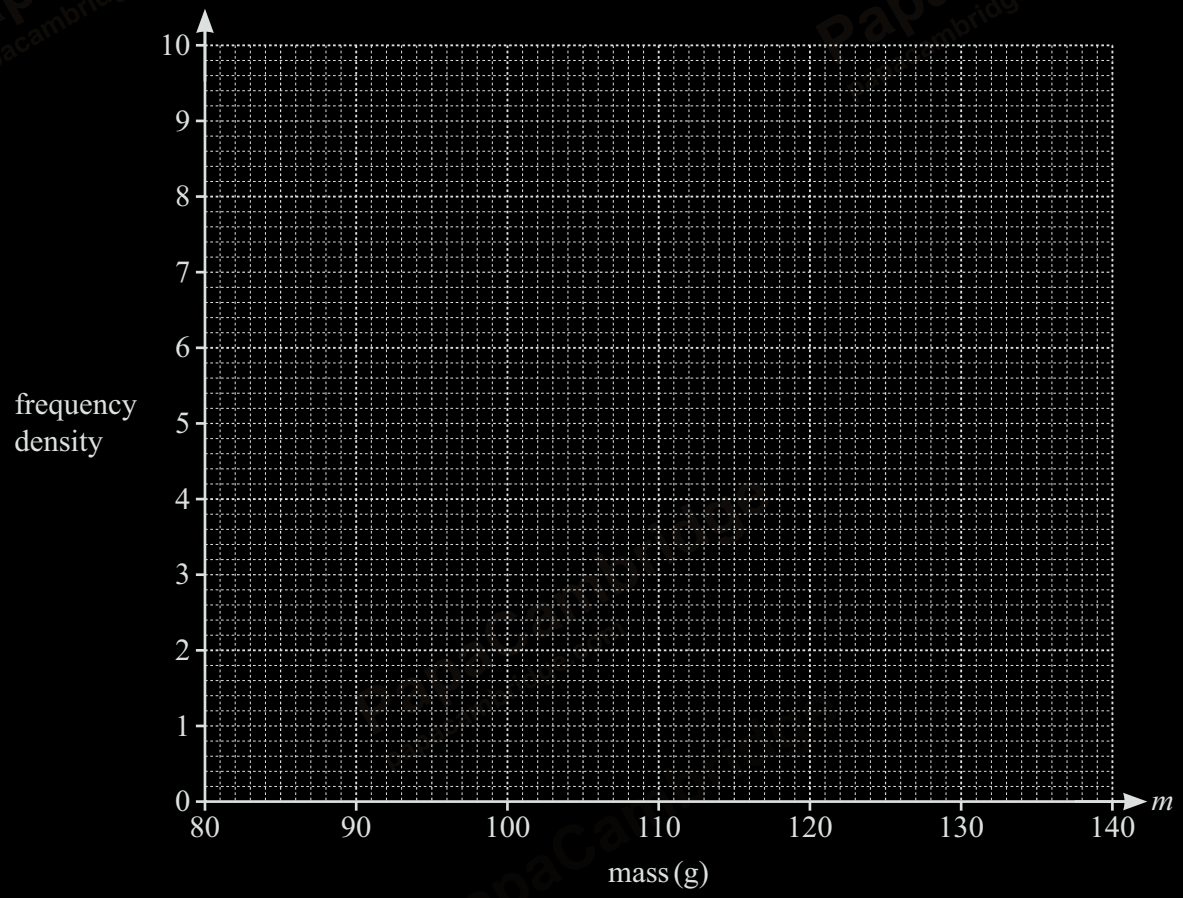
**18** The masses of 200 apples are summarised in the table.

| **mass ($m$ grams)** | $80 < m \leqslant 100$ | $100 < m \leqslant 110$ | $110 < m \leqslant 120$ | $120 < m \leqslant 140$ |
|---|---|---|---|---|
| **frequency** | 30 | 50 | 80 | 40 |

**(a)** Find the percentage of apples with a mass greater than 110 g.

..... % [2]

**(b)** On the grid, draw a histogram to represent the information in the table.



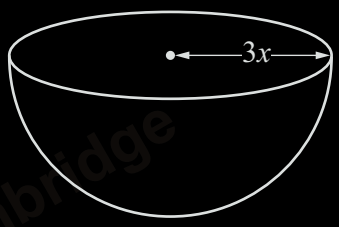

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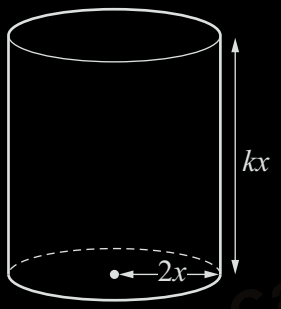
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**19** The diagram shows a solid hemisphere and a solid cylinder.
All the lengths are in centimetres.




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The radius of the hemisphere is $3x$.
The radius of the cylinder is $2x$.
The height of the cylinder is $kx$.

The volume of the hemisphere is equal to the volume of the cylinder.

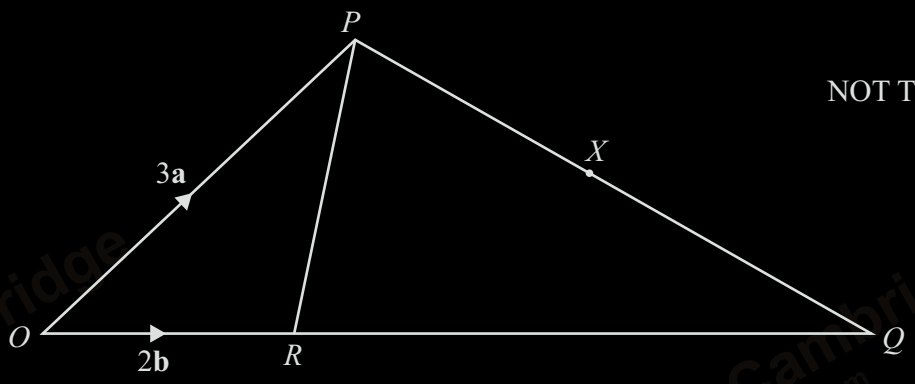
Work out the total surface area of the cylinder.
Give your answer in the form $a\pi x^2$, where $a$ is an integer.

..... $cm^2$ [6]

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**20**




NOT TO SCALE

$OPQ$ is a triangle and $O$ is the origin.
$R$ is a point on $OQ$ and $OQ = 3OR$.
$\overrightarrow{OP} = 3\mathbf{a}$ and $\overrightarrow{OR} = 2\mathbf{b}$.
$X$ is the midpoint of $PQ$.

Find as simply as possible in terms of **a** and/or **b**

**(a)** $\overrightarrow{PR}$

$\overrightarrow{PR} =$ ..... [1]

**(b)** $\overrightarrow{RQ}$

$\overrightarrow{RQ} =$ ..... [1]

**(c)** the position vector of $X$.

..... [3]

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**21** Write as a single fraction in its simplest form.

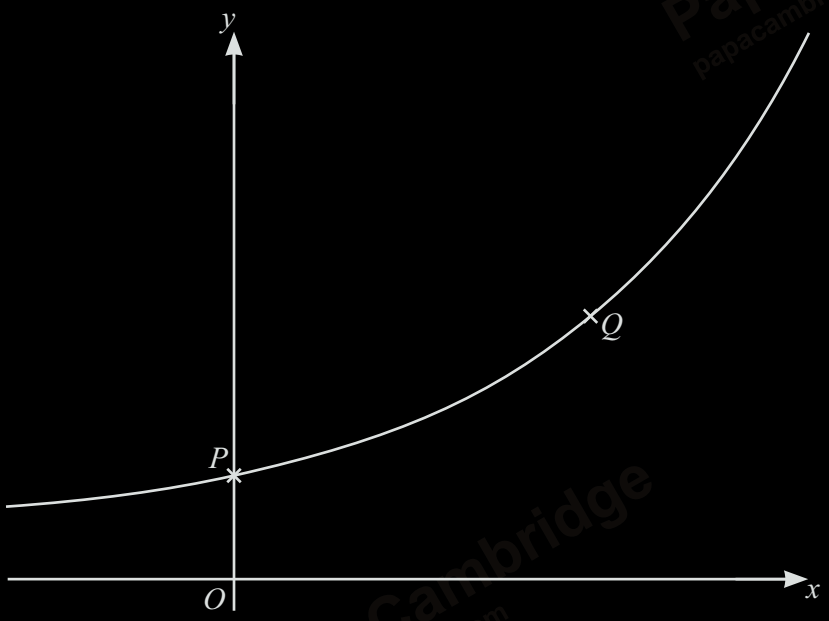
$$\frac{2x+3}{x-1} - \frac{5x}{4-x}$$

..... [4]

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22




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The diagram shows a sketch of the graph of $y = a^x + b$.
The graph passes through point $P$ (0, 5) and point $Q$ (2, 13).

**(a)** Find the value of $a$ and the value of $b$.

$a =$ .....

$b =$ .....

[3]

**(b)** Find the equation of the asymptote to the graph.

..... [1]

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**23** Find the value of $x$.

$$27^{-\frac{4}{3}} = 9^x$$

$x =$ ..... [3]

**24** Expand and simplify.

$$\left(3+5\sqrt{2}\right)\left(2\sqrt{8}-1\right)$$

Give your answer in the form $a+b\sqrt{2}$.

..... [3]

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